

Robust nonlinear projection-based reduced-order models: comparative assessment of closure and manifold strategies

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Abstract

We compare four ANN-enhanced reduced-order modeling routes under aggressive online compression: (i) closure-based intrusive ROMs (PROM-ANN), (ii) latent-manifold intrusive ROMs (PROM-POD-AE), (iii) non-intrusive coefficient-space POD-NN-ROM surrogates $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$, and (iv) non-intrusive latent POD-DL-ROM surrogates $(\boldsymbol{\mu}, t) \mapsto \mathbf{z} \mapsto \mathbf{q}_N$. For intrusive models, the online state is obtained by time-discrete least-squares Petrov–Galerkin (LSPG) minimization, solved by Gauss–Newton, and accelerated by ECSW hyperreduction. The closure family includes state-conditioned, parameter–time conditioned, and hybrid maps for truncated coefficients. The latent family evolves low-dimensional latent coordinates and reconstructs reduced coefficients through a decoder.

A central methodological choice is ROM-consistent training: learning targets are generated by a linear n_{tot} -dimensional HPROM solve in the full truncated space, rather than by direct projection of HDM trajectories. Since closure errors in reconstructed high-order coefficients can feed back into the resolved coordinates of an intrusive LSPG solve, we also introduce a metric-aware basis construction based on an M -regularized LSPG sensitivity metric. This metric is used as a robustness device for the main HPROM-consistent campaign, not as a model-specific tuning parameter. We also consider a second enrichment stage built from additional low-cost HPROM trajectories at Latin-hypercube parameter points. This manuscript consolidates the mathematical formulation and experimental protocol for the 250×250 2D parametric inviscid Burgers benchmark and prepares an extension to a shallow-water benchmark.

1 Introduction

Projection-based model order reduction (PMOR) is a standard framework for reducing high-dimensional parametric models while preserving governing equations [1, 2]. For nonlinear transport-dominated dynamics, linear subspaces may require large dimensions, which motivates nonlinear manifolds, closure modeling, and latent representations [3, 4, 5].

From a machine-learning viewpoint, four families are relevant to this work:

- intrusive closure-enhanced projection (PROM-ANN) [3],
- intrusive latent-manifold projection (PROM-POD-AE), in the spirit of hyperreduced nonlinear manifold ROMs [6],
- non-intrusive direct-coefficient regression maps $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$,
- non-intrusive latent POD-DL-ROM maps [7].

Naming convention used in this manuscript. When we say **POD-NN-ROM**, we refer to the non-intrusive coefficient-space map $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$. When we say **POD-DL-ROM**, we refer to the non-intrusive latent family in the sense of Fresca–Manzoni [7]. When we say **PROM-POD-AE**, we refer to an intrusive latent-manifold projection route (conceptually aligned with intrusive nonlinear-manifold HPROM literature such as [6]), even though those references use different method names.

The goal is a consistent comparison under strong online compression, using the same HDM residual, same time discretization, and the same HPROM intrusive solver/hyperreduction stack for intrusive variants.

2 Linear HPROM Reference and Notation

2.1 Generic semi-discrete IVP and residual notation

Given the generic, parametric, nonlinear, semi-discrete, first-order initial value problem

$$\mathbf{M}(\boldsymbol{\mu}) \dot{\mathbf{u}}(t; \boldsymbol{\mu}) + \mathbf{f}(\mathbf{u}(t; \boldsymbol{\mu}); \boldsymbol{\mu}) - \mathbf{g}(t; \boldsymbol{\mu}) = \mathbf{0}, \quad \mathbf{u}(0; \boldsymbol{\mu}) = \mathbf{u}_0(\boldsymbol{\mu}), \quad (1)$$

with $\boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$, we denote by $\mathbf{r}^{m+1}$ the nonlinear time-discrete residual associated with the selected implicit time integrator. In compact form, each HDM step is written as

$$\mathbf{r}^{m+1}(\mathbf{u}^{m+1}; \boldsymbol{\mu}) = \mathbf{0}. \quad (2)$$

For readability, dependencies on previous time states and Δt are kept implicit in $\mathbf{r}^{m+1}$.

2.2 POD basis from SVD

Let $\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the HDM state with parameters $\boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$. Define

$$\mathbf{u}_{\text{ref}} := \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{u}^i, \quad \mathbf{S}_c = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}} \quad \cdots \quad \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}]. \quad (3)$$

Let

$$\mathbf{S}_c = \mathbf{U}_S \boldsymbol{\Sigma}_S \mathbf{Y}_S^T \quad (4)$$

be its thin SVD, and define the linear ROB

$$\mathbf{V}_{\text{tot}} \in \mathbb{R}^{N \times n_{\text{tot}}} \quad (5)$$

as the first n_{tot} left singular vectors of $\mathbf{U}_S$.

The linear reduced reconstruction is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N(t; \boldsymbol{\mu}), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \in \mathbb{R}^{n_{\text{tot}}}. \quad (6)$$

2.3 Linear time-discrete LSPG reference form

Let $\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the HDM time-discrete residual at step $m+1$. The unsampled reference linear LSPG solve is

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (7)$$

With

$$\mathbf{J}^{m+1,k} := \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}}, \quad \mathbf{T}_N := \mathbf{V}_{\text{tot}}, \quad \mathbf{W}_N^{m+1,k} := \mathbf{J}^{m+1,k} \mathbf{T}_N, \quad (8)$$

the Gauss–Newton increment solves

$$(\mathbf{W}_N^{m+1,k})^T \mathbf{W}_N^{m+1,k} \Delta \mathbf{q}_N^{m+1,k} = -(\mathbf{W}_N^{m+1,k})^T \mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N^{m+1,k}; \boldsymbol{\mu}). \quad (9)$$

2.4 Linear HPROM via ECSW

Assume element-wise residual decomposition

$$\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) = \sum_{e_i \in \mathcal{E}} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ \mathbf{u}; \boldsymbol{\mu}). \quad (10)$$

ECSW approximation yields

$$\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}) = \sum_{e_i \in \tilde{\mathcal{E}}} \xi_{e_i} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ [\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}]; \boldsymbol{\mu}), \quad \xi_{e_i} > 0, \quad (11)$$

with sampled set $\tilde{\mathcal{E}} \subset \mathcal{E}$ [8, 9]. The linear HPROM solve is

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (12)$$

ROM-consistent training targets. All learning targets in this work are extracted from trajectories $\mathbf{q}_N(t; \boldsymbol{\mu})$ generated by the linear HPROM solve (12). They are not obtained by orthogonal projection of HDM snapshots. Thus, the learned maps approximate relations between ROM-generated coordinates that satisfy the same reduced optimality structure used online. This distinction is essential: training on projected HDM coefficients would mix a projection target with an online LSPG/HPROM dynamics, whereas the present protocol keeps data generation, training, and online evaluation consistent. Equation (7) is retained only as the unsampled LSPG reference form.

3 Metric-Aware Basis Construction for Closure Robustness

The closure models considered below depend on a split $\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}]$ between online coordinates $\mathbf{q}$ and reconstructed coordinates $\bar{\mathbf{q}}$. This split is not only an approximation choice: in an intrusive LSPG solve, errors in the reconstructed high-order coordinates can feed back into the low-order coordinates through the residual normal equations. Consequently, the metric used to construct $\mathbf{V}_{\text{tot}}$ becomes part of the robustness design.

3.1 Euclidean and M-regularized LSPG-sensitive metrics

The baseline basis is the usual Euclidean POD basis obtained from the centered snapshot matrix $\mathbf{S}_c$. We also consider a residual-sensitive metric that combines a physical mass contribution with an averaged LSPG sensitivity:

$$\mathbf{W} = \varepsilon \mathbf{M} + \sum_{s \in \mathcal{S}_{\text{met}}} \omega_s (\mathbf{J}_s)^T \mathbf{P}_s \mathbf{J}_s. \quad (13)$$

Here $\mathbf{M}$ is the block mass/area matrix associated with the state inner product, $\mathbf{J}_s = \partial \mathbf{r}_s / \partial \mathbf{u}$ is the time-discrete residual Jacobian at a selected snapshot, $\mathbf{P}_s$ is the nonnegative sampling/weighting operator used for the residual metric, and ω_s are quadrature and parameter weights. The small positive parameter ε regularizes the metric and preserves physical state weighting when the residual sensitivity is weak or locally rank deficient.

The corresponding weighted POD basis is obtained from the generalized correlation matrix

$$\mathbf{C} = \mathbf{S}_c^T \mathbf{W} \mathbf{S}_c. \quad (14)$$

If $\mathbf{C} \mathbf{y}_i = \lambda_i \mathbf{y}_i$, the associated modes are

$$\phi_i = \frac{1}{\sqrt{\lambda_i}} \mathbf{S}_c \mathbf{y}_i, \quad \phi_i^T \mathbf{W} \phi_j = \delta_{ij}. \quad (15)$$

In practice, the metric is assembled on a representative subset of parameter and time samples, selected independently from the final test points. The resulting basis is then used uniformly by all reduced models; it is not tuned separately for a particular closure map.

3.2 Low/high transfer diagnostic

To quantify why such a metric may matter for Case 2, consider a local linearization of the LSPG normal equations around a reference reduced trajectory. Let

$$\mathbf{A}_s = (\mathbf{J}_s)^T \mathbf{P}_s \mathbf{J}_s, \quad \mathbf{H}_s = \begin{bmatrix} \mathbf{V}^T \mathbf{A}_s \mathbf{V} & \mathbf{V}^T \mathbf{A}_s \bar{\mathbf{V}} \\ \bar{\mathbf{V}}^T \mathbf{A}_s \mathbf{V} & \bar{\mathbf{V}}^T \mathbf{A}_s \bar{\mathbf{V}} \end{bmatrix}. \quad (16)$$

If the high coordinates are perturbed by $\delta \bar{\mathbf{q}}$, the corresponding first-order low-coordinate response is approximated by

$$\delta \mathbf{q} \approx -(\mathbf{H}_{LL})^{-1} \mathbf{H}_{LH} \delta \bar{\mathbf{q}} = \mathbf{T}_{LH} \delta \bar{\mathbf{q}}. \quad (17)$$

Thus, $\mathbf{T}_{LH}$ measures which high-coordinate directions are most dangerous for the resolved LSPG coordinates. The purpose of the M -regularized LSPG-sensitive metric is not to optimize a single regression model, but to construct a trial basis for which this low/high transfer is less amplifying while preserving the accuracy of the remaining model families.

3.3 When the metric is needed

Table 1 summarizes the role of the metric for the model classes considered in this paper. The guiding principle is conservative: the LSPG-sensitive basis is justified only if it improves the robustness of closure-based intrusive models without degrading the linear reference, PROM-ANN Cases 1 and 3, PROM-POD-AE, or the non-intrusive baselines.

Table 1: Role of the basis metric in each model family.

Model family	Why the metric may matter	Practical criterion
Linear HPROM	Changes the reduced trial space and therefore the reference trajectory.	Must not degrade the baseline projection/HPROM accuracy appreciably.
PROM-ANN Case 1, $\bar{\mathbf{q}} = \mathcal{N}(\mathbf{q})$	Closure errors are tied to the online low coordinates.	Check that the modified coordinates remain learnable and stable.
PROM-ANN Case 2, $\bar{\mathbf{q}} = \mathcal{M}(\boldsymbol{\mu}, t)$	High coordinates are injected independently of the online solve.	Most sensitive case; reduce low/high transfer in (17).
PROM-ANN Case 3, $\bar{\mathbf{q}} = \mathcal{H}(\mathbf{q}, \boldsymbol{\mu}, t)$	Hybrid inputs may compensate part of the closure error.	Verify that robustness improves without increasing online stiffness.
PROM-POD-AE	The decoder acts on coordinates defined by the selected basis.	Preserve latent reconstruction quality and LSPG convergence.
POD-NN-ROM/POD-DL-ROM	No intrusive feedback, but the regression targets change.	Assess coefficient and state errors; do not optimize only for Case 2.

The numerical evidence used to select the M -regularized LSPG-sensitive basis for the Burgers campaign is reported later, after the benchmark has been introduced, in Appendix A. It consists of the direct low/high transfer measure in (17) and an oracle high-mode perturbation check. Unless explicitly stated otherwise, the subsequent HPROM-consistent Burgers pipeline uses this LSPG-sensitive basis as the main reduced trial basis.

4 Method I: PROM-ANN Closure Models

4.1 Manifold definitions

For closure models, we introduce the partition

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \bar{\mathbf{V}}], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}, \quad n + \bar{n} = n_{\text{tot}}. \quad (18)$$

PROM-ANN uses retained coordinates $\mathbf{q} \in \mathbb{R}^n$ as online unknowns and reconstructs truncated coordinates via learned maps:

$$\text{Case 1: } \bar{\mathbf{q}} = \mathcal{N}(\mathbf{q}), \quad (19)$$

$$\text{Case 2: } \bar{\mathbf{q}} = \mathcal{M}(\boldsymbol{\mu}, t), \quad (20)$$

$$\text{Case 3: } \bar{\mathbf{q}} = \mathcal{H}(\mathbf{q}, \boldsymbol{\mu}, t). \quad (21)$$

In the n_{tot} -dimensional linear HPROM used for target generation, both $\mathbf{q}$ and $\bar{\mathbf{q}}$ are solved online. In the PROM-ANN models, only $\mathbf{q}$ remains an online unknown; the secondary coordinates are injected by the learned closure. This gives an n -dimensional nonlinear trial manifold embedded in the same n_{tot} -dimensional reduced space used to generate the training data.

The state approximation is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}(t; \boldsymbol{\mu}) + \bar{\mathbf{V}} \mathcal{F}(\mathbf{q}, \boldsymbol{\mu}, t), \quad (22)$$

with $\mathcal{F} \in \{\mathcal{N}, \mathcal{M}, \mathcal{H}\}$.

The three cases differ in how the closure interacts with the LSPG tangent. In Case 1 and Case 3 the tangent contains $\bar{\mathbf{V}} \partial \mathcal{F} / \partial \mathbf{q}$, whereas in Case 2 the closure is parameter-time conditioned and the tangent with respect to the online coordinate is simply $\mathbf{V}$. Case 2 is therefore the cheapest intrusive closure, but it is also the most exposed to high-coordinate injection errors because the injected block is not corrected through a state-dependent tangent.

4.2 LSPG/Gauss–Newton solve

At each time step,

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}); \boldsymbol{\mu})\|_2^2. \quad (23)$$

Define

$$\mathbf{J}^{m+1,k} := \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}}, \quad \mathbf{T}_q^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}}, \quad \mathbf{W}_q^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_q^{m+1,k}. \quad (24)$$

The Gauss–Newton increment solves

$$(\mathbf{W}_q^{m+1,k})^T \mathbf{W}_q^{m+1,k} \Delta \mathbf{q}^{m+1,k} = -(\mathbf{W}_q^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (25)$$

4.3 ECSW hyperreduced form

PROM-ANN HPROM solves

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2, \quad (26)$$

with test basis $\widetilde{\mathbf{W}}_q = \widetilde{\mathbf{J}} \mathbf{T}_q$.

Offline training targets. For Cases 1–3, supervised data are extracted from ROM-consistent trajectories $\mathbf{q}_N = [\mathbf{q}; \bar{\mathbf{q}}]$. Case 1 trains $\mathcal{N}$ from $(\mathbf{q}, \bar{\mathbf{q}})$, Case 2 trains $\mathcal{M}$ from $(\boldsymbol{\mu}, t, \bar{\mathbf{q}})$, and Case 3 trains $\mathcal{H}$ from $(\mathbf{q}, \boldsymbol{\mu}, t, \bar{\mathbf{q}})$.

5 Method II: PROM-POD-AE Latent Manifold

5.1 Autoencoder and latent manifold

Let

$$\mathbf{z} = \mathcal{E}(\mathbf{q}_N), \quad \hat{\mathbf{q}}_N = \mathcal{D}(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^{n_z}, \quad n_z \ll n_{\text{tot}}. \quad (27)$$

The intrusive latent manifold is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z}(t; \boldsymbol{\mu})). \quad (28)$$

This is an intrusive latent-manifold projection approach, in line with intrusive nonlinear-manifold HPROM literature [6], and is separated from the non-intrusive POD-DL-ROM line [7].

Offline autoencoder training. Given ROM-consistent targets $\mathbf{q}_N^{\text{ref}}$, the autoencoder $(\mathcal{E}, \mathcal{D})$ is trained by reconstruction in reduced coordinates:

$$\mathcal{L}_{\text{AE}} = \frac{1}{N_{\text{samp}}} \sum_{j=1}^{N_{\text{samp}}} \left\| \mathbf{q}_{N,j}^{\text{ref}} - \mathcal{D}(\mathcal{E}(\mathbf{q}_{N,j}^{\text{ref}})) \right\|_2^2. \quad (29)$$

The resulting decoder $\mathcal{D}$ defines the nonlinear trial manifold used in the intrusive online LSPG solve (30).

5.2 LSPG/Gauss–Newton solve in latent coordinates

The online unknown is $\mathbf{z}^{m+1} \in \mathbb{R}^{n_z}$:

$$\mathbf{z}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{y} \in \mathbb{R}^{n_z}} \left\| \mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{y}); \boldsymbol{\mu}) \right\|_2^2. \quad (30)$$

Define

$$\mathbf{T}_z^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{z}} = \mathbf{V}_{\text{tot}} \frac{\partial \mathcal{D}}{\partial \mathbf{z}}, \quad \mathbf{W}_z^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_z^{m+1,k}. \quad (31)$$

The Gauss–Newton step solves

$$(\mathbf{W}_z^{m+1,k})^T \mathbf{W}_z^{m+1,k} \Delta \mathbf{z}^{m+1,k} = -(\mathbf{W}_z^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{z}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (32)$$

5.3 ECSW hyperreduced form

PROM-POD-AE HPROM solves

$$\mathbf{z}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{y} \in \mathbb{R}^{n_z}} \left\| \tilde{\mathbf{r}}^{m+1}(\mathbf{y}; \boldsymbol{\mu}) \right\|_2^2, \quad (33)$$

with test basis $\tilde{\mathbf{W}}_z = \tilde{\mathbf{J}} \mathbf{T}_z$.

6 Method III: Non-Intrusive POD-NN-ROM

Model definition. The non-intrusive baseline maps exogenous inputs directly to reduced coefficients:

$$\mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{G}_q(\boldsymbol{\mu}, t). \quad (34)$$

This model does not require online residual minimization.

Training objective and online cost. Given ROM-consistent targets $\mathbf{q}_N^{\text{ref}}(t; \boldsymbol{\mu})$, one minimizes a supervised loss (e.g., MSE)

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{samp}}} \sum_{j=1}^{N_{\text{samp}}} \left\| \mathcal{G}_q(\boldsymbol{\mu}_j, t_j) - \mathbf{q}_{N,j}^{\text{ref}} \right\|_2^2. \quad (35)$$

Online prediction is then a direct forward evaluation of $\mathcal{G}_q$ followed by reconstruction with $\mathbf{V}_{\text{tot}}$.

7 Method IV: POD-DL-ROM (Latent, Non-Intrusive)

Model definition. The second non-intrusive baseline follows the POD-DL-ROM family [7]:

$$\mathbf{z}(t; \boldsymbol{\mu}) \approx \mathcal{G}_z(\boldsymbol{\mu}, t), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)), \quad (36)$$

where $\mathcal{D}$ is a decoder trained offline and $\mathbf{z} \in \mathbb{R}^{n_z}$.

Training objective and online cost. In the present formulation, the autoencoder block $(\mathcal{E}, \mathcal{D})$ follows the same training strategy and normalization used in Section 5: reconstruction of ROM-consistent $\mathbf{q}_N$ snapshots through a latent bottleneck. The POD-DL-ROM dynamics map then replaces the POD-NN-ROM output variable $\mathbf{q}_N$ by $\mathbf{z}$, i.e., it learns $(\boldsymbol{\mu}, t) \mapsto \mathbf{z}$ and composes with the same decoder $\mathcal{D}$:

$$\mathbf{q}_N(\boldsymbol{\mu}, t) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)). \quad (37)$$

Hence, POD-NN-ROM and POD-DL-ROM share the same exogenous-input principle $(\boldsymbol{\mu}, t) \mapsto (\cdot)$, with POD-DL-ROM acting in latent coordinates as proposed in [7]. Online evaluation remains non-intrusive (single forward pass) and does not involve residual minimization.

8 Physics-Regularized Variant (Ancillary)

Physics-regularized neural training adds a residual-based term to a supervised objective [10, 11]. In this study, such variants are treated as optional diagnostics (not core baselines) because they remain significantly more expensive offline and do not consistently improve accuracy in the present benchmark.

9 Burgers Benchmark and Experimental Protocol

9.1 Governing equations and parameter domain

We use the 2D parametric inviscid Burgers benchmark from [3], in conservative vector form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad (38)$$

with

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2}u_x^2 \\ \frac{1}{2}u_x u_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2}u_x u_y \\ \frac{1}{2}u_y^2 \end{bmatrix}, \quad \mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}. \quad (39)$$

The problem is defined on $\Omega = [0, 100] \times [0, 100]$ with final time $T_f = 25$ and parameters $\boldsymbol{\mu} = (\mu_1, \mu_2) \in \mathcal{D}$, where

$$\mathcal{D} = [4.25, 5.50] \times [0.015, 0.03]. \quad (40)$$

This conservative form is the one used by the discrete residual and Jacobian in the ROM/HPROM pipeline. The initial/boundary conditions used in this benchmark are

$$u_x(0, y, t; \boldsymbol{\mu}) = \mu_1, \quad u_x(x, y, 0) = 1, \quad u_y(x, y, 0) = 1. \quad (41)$$

9.2 Numerical discretization

The HDM uses a cell-centered finite-volume Godunov discretization on a 250×250 mesh, hence $N = 2N_x N_y = 125,000$. Time integration uses 500 uniform steps with $\Delta t = 0.05$ (thus 501 stored states including $t = 0$). The same time-discrete residual is used by linear HPROM and by all intrusive ANN-enhanced variants (PROM-ANN and PROM-POD-AE) through LSPG/Gauss–Newton solves.

9.3 Training and evaluation protocol

Stage 1 builds $\mathbf{V}_{\text{tot}}$ from HDM snapshots at 9 structured training parameters (a 3×3 tensor grid on $\mathcal{D}$), giving $N_s = 9 \times 501 = 4509$ snapshots. Stage 2 generates ROM-consistent coefficients $\mathbf{q}_N(t; \boldsymbol{\mu})$ by solving the linear full-space reduced model (n_{tot}). Stage 3 trains the learned maps from these reduced targets.

Primary evaluation is reported at one verification point and two off-grid points:

$$\boldsymbol{\mu}^{(v)} = (4.875, 0.0225), \quad \boldsymbol{\mu}^{(1)} = (4.56, 0.019), \quad \boldsymbol{\mu}^{(2)} = (5.19, 0.026). \quad (42)$$

For the expanded-enrichment study, we also add an extrapolatory stress-test point located outside the original training box,

$$\boldsymbol{\mu}^{(3)} = (4.00, 0.0330), \quad (43)$$

which is 20% beyond the upper-left corner of $\mathcal{D}$ in both parameter directions. Throughout the tables and figures, the superscript (v) denotes the verification point, (1) and (2) denote the two off-grid test points, and (3) denotes the extrapolatory stress-test point. Results are presented in the order $\boldsymbol{\mu}^{(v)}$, $\boldsymbol{\mu}^{(1)}$, $\boldsymbol{\mu}^{(2)}$, and $\boldsymbol{\mu}^{(3)}$ in the main baseline and enriched result tables and figures.

In addition to the baseline 3×3 training set, we consider an expanded enriched campaign with 36 Latin-hypercube points generated by the fixed linear HPROM. The enrichment contains 18 points inside the original domain $\mathcal{D}$ and 18 points in a 25% expanded parameter box,

$$[3.9375, 5.8125] \times [0.01125, 0.03375]. \quad (44)$$

No additional HDM trajectories are computed. The enriched Stage 2 dataset therefore contains 45 ROM-consistent trajectories, or $45 \times 501 = 22,545$ coefficient samples. The enrichment run keeps the same MLSPG-sensitive basis and reuses the fixed Stage-2 linear ECSW weights from the baseline campaign; the linear ECSW rule is not copied, rebuilt, or retuned. The four evaluation points are excluded from the enrichment set. Figures 1 and 2 summarize the baseline and expanded enriched sampling sets.

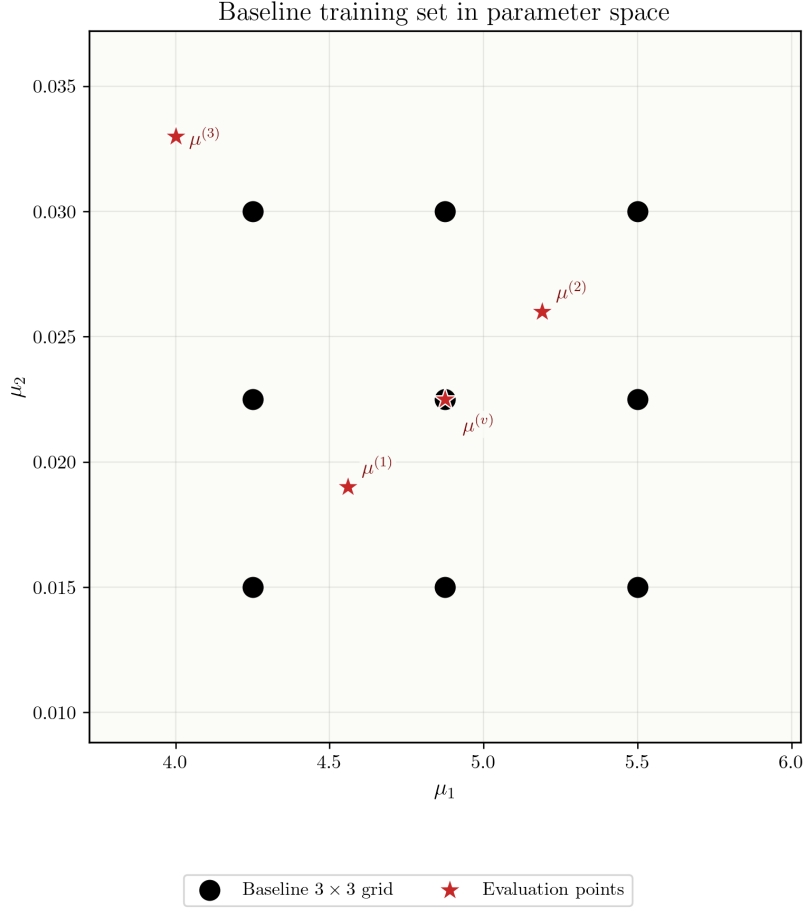


Figure 1: Baseline training set and evaluation points in parameter space. The axes are shared with Figure 2 so that the location of the extrapolatory stress-test point $\mu^{(3)}$ is shown on the same scale as the expanded enrichment.

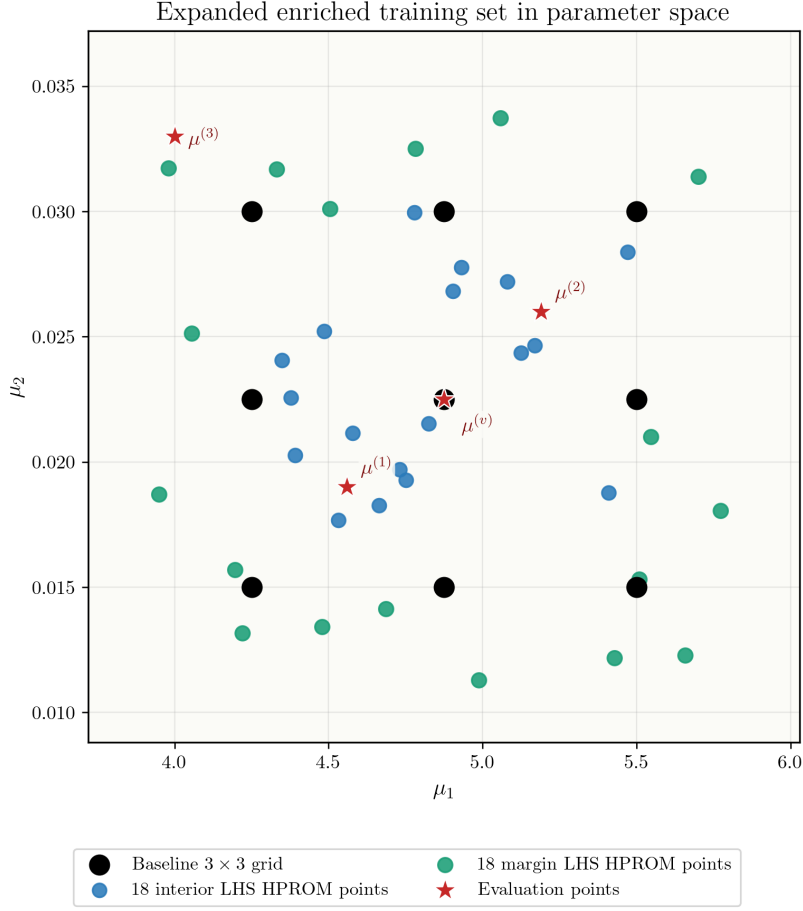


Figure 2: Expanded enriched MLSPG-sensitive campaign: baseline 3×3 grid, 18 interior LHS points, 18 margin LHS points generated with the fixed linear HPRM, and the four evaluation points. The axes match Figure 1; the margin points lie in the 25% expanded sampling box around the original domain $\mathcal{D}$.

9.4 HPRM-Consistent Evaluation Pipeline

To avoid protocol-induced bias in intrusive comparisons, we enforce one end-to-end HPRM pipeline: Stage 2 reduced targets are generated with linear HPRM in $\mathbf{V}_{\text{tot}}$, Stage 3 intrusive models are trained on these HPRM-consistent reduced trajectories, and online intrusive evaluations are performed with linear HPRM settings. This applies uniformly to Linear HPRM, PROM-ANN (Cases 1–3), and PROM-POD-AE.

9.5 Reported Metrics

For each evaluation parameter, we report complementary state-space, coefficient-space, and run-time metrics. State accuracy is measured by the relative space-time trajectory error versus HDM (defined in Section 10, Error Definitions). Coefficient-space comparisons are reported against the linear n_{tot} reference, including per-index absolute/relative modal errors and time-resolved modal error heatmaps. Computational performance is reported through online runtimes and speedups relative to the HDM mean runtime.

10 Burgers Results in the HPRoM-Consistent Setting

This section reports the Burgers results for the completed model classes and configurations considered here: the linear HPRoM reference, PROM-ANN Cases 1–3, intrusive PROM-POD-AE, non-intrusive POD-NN-ROM $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$, and non-intrusive POD-DL-ROM.

10.1 Error Definitions

The trajectory error reported in tables and summaries is the relative space-time Frobenius error (in %):

$$\varepsilon_u^{(m)}(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{U}_{\text{HDM}}(\boldsymbol{\mu}) - \mathbf{U}_m(\boldsymbol{\mu})\|_F}{\|\mathbf{U}_{\text{HDM}}(\boldsymbol{\mu})\|_F}, \quad (45)$$

where $\mathbf{U}_{\text{HDM}}, \mathbf{U}_m \in \mathbb{R}^{N \times N_t}$ collect all states along time.

Coefficient diagnostics use the linear n_{tot} -HPRoM coefficients $\mathbf{Q}_{\text{ref}} = [\mathbf{q}_{\text{ref}}(t_1) \cdots \mathbf{q}_{\text{ref}}(t_{N_t})]$ as reference and model coefficients $\mathbf{Q}_m$:

$$\varepsilon_{q_i, \text{abs}}^{(m)} = \|q_{\text{ref}, i}(\cdot) - q_{m, i}(\cdot)\|_2, \quad \varepsilon_{q_i, \text{rel}}^{(m)} = \frac{\|q_{\text{ref}, i}(\cdot) - q_{m, i}(\cdot)\|_2}{\|q_{\text{ref}, i}(\cdot)\|_2}, \quad (46)$$

and heatmaps display

$$e_{i, k}^{(m)} = |q_{\text{ref}, i}(t_k) - q_{m, i}(t_k)|, \quad \hat{e}_{i, k}^{(m)} = \frac{|q_{\text{ref}, i}(t_k) - q_{m, i}(t_k)|}{\|q_{\text{ref}, i}(\cdot)\|_2}. \quad (47)$$

For online cost tables, intrusive methods are timed with their online Gauss–Newton solve, while non-intrusive methods are timed through pure network inference. Speedup is defined with respect to the mean HDM runtime per parameter,

$$S^{(m)} = \frac{t_{\text{HDM}}}{t_m}, \quad (48)$$

where the current campaign uses $t_{\text{HDM}} = 737.44$ s, obtained from the mean runtime over the nine HDM training trajectories.

10.2 Current MLSPG-Sensitive HPRoM Campaign

The first production campaign uses the M -regularized LSPG-sensitive basis introduced above and the HPRoM backend generated from this basis. The completed non-enriched campaign contains the linear HPRoM, PROM-ANN Cases 1–3, intrusive PROM-POD-AE, non-intrusive POD-NN-ROM, and non-intrusive POD-DL-ROM. Every reported state and coefficient trajectory was generated with the same MLSPG-sensitive basis and the same Stage 2 linear-HPRoM training dataset; no result from the earlier Euclidean/PROM exploratory campaign is included.

Table 2 reports the Stage-3 networks selected for the non-enriched campaign in reader-facing form. The PROM-ANN and POD-NN-ROM rows use the same feed-forward MLP widths, while PROM-POD-AE and POD-DL-ROM use explicit latent bottlenecks with $n_z = 10$.

Table 2: Stage-3 network architectures used in the non-enriched MLSPG-sensitive campaign. The table reports the learned map, hidden widths or latent dimensions, normalization/activation choices, validation relative Frobenius error on the HPROM coefficient dataset, and the number of trainable parameters.

Model	Learned map	Network architecture	Normalization / activation	Val. rel. Frobenius (%)	Trainable params
PROM-ANN Case 1	$\mathbf{q} \in \mathbb{R}^{10} \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{141}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	1.371	564 621
PROM-ANN Case 2 ($n = 10$)	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{141}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	2.337	562 829
PROM-ANN Case 2 ($n = 20$)	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{131}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	4.550	560 259
PROM-ANN Case 3	$(\mathbf{q}, \mu_1, \mu_2, t) \in \mathbb{R}^{13} \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{141}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	0.537	565 389
PROM-POD-AE	$\mathbf{q}_N \in \mathbb{R}^{151} \mapsto \mathbf{z} \in \mathbb{R}^{10} \mapsto \bar{\mathbf{q}}_N$	Encoder (512, 256, 128); decoder reverse; $n_z = 10$	z-score; GELU	0.131	486 817
POD-NN-ROM	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \mathbf{q}_N \in \mathbb{R}^{151}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	0.908	565 399
POD-DL-ROM	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \mathbf{z} \in \mathbb{R}^{10} \mapsto \bar{\mathbf{q}}_N$	Encoder (512, 256); decoder (256, 512); dynamics (256, 512, 512, 256); $n_z = 10$	z-score; SiLU	0.607	952 747

Table 3: Current MLSPG-sensitive campaign: relative trajectory errors (%) with respect to HDM. The first three rows are the original verification/off-grid evaluation points; the final row is the extrapolatory stress-test point. Intrusive learned models use 2% ECSW rules; POD-NN-ROM and POD-DL-ROM are non-intrusive.

Point	Linear HPROM	PROM-ANN Case 1	PROM-ANN Case 2 ($n = 10$)	PROM-ANN Case 2 ($n = 20$)	PROM-ANN Case 3	PROM-POD-AE	POD-NN-ROM	POD-DL-ROM
$\mu^{(v)}$ (verification)	0.413	0.424	0.437	0.441	0.415	0.458	0.479	0.439
$\mu^{(1)}$ (off-grid)	0.460	0.600	1.808	3.433	0.521	0.633	1.927	1.504
$\mu^{(2)}$ (off-grid)	0.472	0.722	1.565	5.569	0.608	0.689	2.396	1.540
$\mu^{(3)}$ (20% extrapolation)	0.868	1.563	4.313	6.440	1.438	1.943	9.192	4.980

On the original verification/off-grid points $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$, the 2% ECSW learned intrusive rows remain close to the fixed linear HPROM for Case 1, Case 3, and PROM-POD-AE. PROM-ANN Case 3 gives errors between 0.415% and 0.608%, Case 1 gives 0.424%–0.722%, and PROM-POD-AE gives 0.458%–0.689%, compared with the linear-HPROM range 0.413%–0.472%. By contrast, the parameter-time Case 2 closures remain more fragile off grid. Case 2 with $n = 10$ reaches 1.808% and 1.565% on the two off-grid points, while Case 2 with $n = 20$ reaches 3.433% and 5.569%. Thus, increasing the number of resolved primary coordinates does not, by itself, improve the parameter-to-tail closure. Appendix B reports an additional diagnostic in which the stronger $n = 10$ Case 2 map is trimmed and reused inside an $n = 20$ online solve, to separate the effect of the online split from the effect of independently training the $n = 20$ closure. Among the non-intrusive models, POD-DL-ROM improves on POD-NN-ROM at both off-grid points and remains comparable at the verification point.

The extrapolatory point $\mu^{(3)}$ is intentionally harder and is not included in the original-three mean used in the enrichment comparison. At this point the linear HPROM remains below 0.9%, while the learned models trained only on the original coefficient set degrade more strongly: PROM-ANN Case 1 reaches 1.563%, Case 2 reaches 4.313% for $n = 10$ and 6.440% for $n = 20$, Case 3 reaches 1.438%, PROM-POD-AE reaches 1.943%, POD-NN-ROM reaches 9.192%, and POD-DL-ROM reaches 4.980%. This behavior motivates the expanded HPROM-generated coefficient enrichment reported below.

To separate the sources of error in the learned intrusive rows, we also run a diagnostic that compares three levels for the same trained nonlinear representations: direct reconstruction from the saved linear-HPROM coefficient trajectory, a full-residual PROM solve without ECSW sampling, and the production 2% ECSW HPROM solve. This diagnostic is not a replacement for the production tables, because the reconstruction-only rows do not solve the online residual. However, it is useful for identifying whether a loss of accuracy is already present in the learned reconstruction map, appears when the reduced coordinates are advanced by the nonlinear PROM dynamics, or is introduced by ECSW sampling.

Table 4: Baseline MLSPG-sensitive intrusive nonlinear models: diagnostic separation of reconstruction, PROM, and HPROM errors. “Recon.” injects the saved linear-HPROM coefficient trajectory into the learned nonlinear reconstruction and performs no online solve; it is a no-feedback diagnostic, not a deployable model. For the closure rows it approximates the error floor associated with the learned high-coordinate map at the teacher coefficients; for PROM-POD-AE it is not necessarily a lower bound because the online latent solve can select latent coordinates different from the encoded teacher trajectory. PROM denotes the full-residual unsampled nonlinear solve using the same learned representation, and HPROM denotes the production 2% ECSW solve. All entries are relative trajectory errors (%) with respect to HDM.

Model	Point	Recon.	PROM	HPROM	PROM-Recon.	HPROM-PROM
PROM-ANN Case 1	$\mu^{(v)}$	0.421	0.412	0.424	-0.009	0.012
PROM-ANN Case 1	$\mu^{(1)}$	0.499	0.510	0.600	0.011	0.090
PROM-ANN Case 1	$\mu^{(2)}$	0.581	0.671	0.722	0.090	0.051
PROM-ANN Case 1	$\mu^{(3)}$	1.132	1.568	1.563	0.436	-0.005
PROM-ANN Case 2 ($n = 10$)	$\mu^{(v)}$	0.433	0.432	0.437	-0.001	0.005
PROM-ANN Case 2 ($n = 10$)	$\mu^{(1)}$	1.100	1.353	1.808	0.253	0.456
PROM-ANN Case 2 ($n = 10$)	$\mu^{(2)}$	1.285	1.352	1.565	0.067	0.213
PROM-ANN Case 2 ($n = 10$)	$\mu^{(3)}$	2.721	2.868	4.313	0.147	1.445
PROM-ANN Case 2 ($n = 20$)	$\mu^{(v)}$	0.440	0.439	0.441	-0.001	0.002
PROM-ANN Case 2 ($n = 20$)	$\mu^{(1)}$	3.155	4.055	3.433	0.900	-0.623
PROM-ANN Case 2 ($n = 20$)	$\mu^{(2)}$	3.240	4.087	5.569	0.848	1.482
PROM-ANN Case 2 ($n = 20$)	$\mu^{(3)}$	3.835	4.997	6.440	1.162	1.443
PROM-ANN Case 3	$\mu^{(v)}$	0.414	0.405	0.415	-0.010	0.010
PROM-ANN Case 3	$\mu^{(1)}$	0.495	0.519	0.521	0.024	0.002
PROM-ANN Case 3	$\mu^{(2)}$	0.565	0.568	0.608	0.003	0.040
PROM-ANN Case 3	$\mu^{(3)}$	1.085	1.385	1.438	0.301	0.053
PROM-POD-AE	$\mu^{(v)}$	0.414	0.406	0.458	-0.008	0.052
PROM-POD-AE	$\mu^{(1)}$	1.919	0.555	0.633	-1.365	0.079
PROM-POD-AE	$\mu^{(2)}$	1.735	0.658	0.689	-1.077	0.031
PROM-POD-AE	$\mu^{(3)}$	3.990	1.926	1.943	-2.064	0.017

Table 5: Aggregate form of Table 4. The first three columns average the original verification/off-grid points $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$; the final three columns report the extrapolatory point $\mu^{(3)}$ separately.

Model	Original three-point mean			$\mu^{(3)}$ extrapolation		
	Recon.	PROM	HPROM	Recon.	PROM	HPROM
PROM-ANN Case 1	0.500	0.531	0.582	1.132	1.568	1.563
PROM-ANN Case 2 ($n = 10$)	0.939	1.046	1.270	2.721	2.868	4.313
PROM-ANN Case 2 ($n = 20$)	2.278	2.861	3.148	3.835	4.997	6.440
PROM-ANN Case 3	0.492	0.497	0.515	1.085	1.385	1.438
PROM-POD-AE	1.356	0.540	0.594	3.990	1.926	1.943

Table 5 shows that PROM-ANN Cases 1 and 3 have small reconstruction floors on the original three points, and their full PROM and HPROM errors remain close to those floors. Case 2 is different: a substantial part of the error is already present in the parameter-time high-coordinate prediction, and the HPROM sampling adds a further penalty, especially at $\mu^{(3)}$. PROM-POD-AE should be read separately: encoding and decoding the linear-HPROM coefficient trajectory is not its best latent representation, so the reconstruction-only row is not

a lower bound. Its full PROM solve finds substantially better latent coordinates and the 2% HPROM row stays close to that full-PROM value.

The hyperreduction information for the same campaign is reported in Table 6. The linear HPROM uses the shared Stage 2 ECSW weights, while the intrusive PROM-ANN models use case-specific ECSW weights trained for their corresponding nonlinear residuals. PROM-POD-AE uses its own latent-manifold ECSW rule, constructed with the decoder tangent entering the reduced residual. For the intrusive learned models, the ECSW training set is not a contiguous block of time steps. It is selected by a parameter–time stratified sampling rule: all available training parameter trajectories are represented and the sampled time indices are spread over the transient, with a shifted time grid used to avoid repeatedly selecting the same temporal locations. The nominal sampling fraction therefore controls the size of the residual/tangent training set used to identify the element weights; the final online mesh size N_e is the number of elements retained with nonzero ECSW weights after the nonnegative least squares solve.

The reported learned intrusive rows use 2% case-specific ECSW rules. This choice is used consistently for PROM-ANN Cases 1–3 and PROM-POD-AE in both the baseline and enriched campaigns. The linear HPROM is not rebuilt for this comparison; it uses the fixed Stage 2 ECSW rule that defines the coefficient teacher. POD-NN-ROM and POD-DL-ROM are included for timing comparison but are non-intrusive and therefore do not require ECSW.

Table 6: Current MLSPG-sensitive campaign: online dimensions, mesh sizes, average online timings over the four reported evaluation points, and speedups with respect to the HDM mean time $t_{\text{HDM}} = 737.44$ s. Learned intrusive models use selected 2% ECSW rules, while the linear HPROM uses the shared Stage-2 rule. POD-NN-ROM and POD-DL-ROM are non-intrusive. $N_e = 62\,500$ is the full number of finite elements.

Method	Online/latent dim.	n_s	n_e	Mean online time (s)	Mean speedup vs HDM
Linear HPROM	151	–	5975	100.6080	7.3
PROM-ANN Case 1	10	141	901	9.0885	81.1
PROM-ANN Case 2 ($n = 10$)	10	141	901	2.6251	280.9
PROM-ANN Case 2 ($n = 20$)	20	131	1801	3.9693	185.8
PROM-ANN Case 3	10	141	901	9.0382	81.6
PROM-POD-AE	10	–	901	8.5749	86.0
POD-NN-ROM	151	–	–	0.0322	22921.2
POD-DL-ROM	10	–	–	0.0206	35757.3

The average online timings in Table 6 include the four reported evaluation points. PROM-POD-AE retains 901 elements and averages 8.57 s per trajectory, corresponding to an $86.0\times$ speedup over HDM. PROM-ANN Case 2 with $n = 10$ is the fastest intrusive model at 2.63 s ($280.9\times$), while Case 2 with $n = 20$ retains 1801 elements and averages 3.97 s ($185.8\times$). Case 1 and Case 3 both average about 9.1 s, with speedups near $81\times$. The non-intrusive online costs are 0.0322 s for POD-NN-ROM and 0.0206 s for POD-DL-ROM. These online speedups do not include the common offline costs of HDM snapshot generation, linear-HPROM target generation, or network training.

10.3 Solution Overlays

Figure 3 compares the HDM, the linear HPROM, all selected intrusive learned HPROMs, and both non-intrusive baselines at the four reported evaluation points. Solid curves correspond to the final time, while dashed curves show intermediate times. The line cuts are the horizontal and vertical midline cuts of the u_x component.

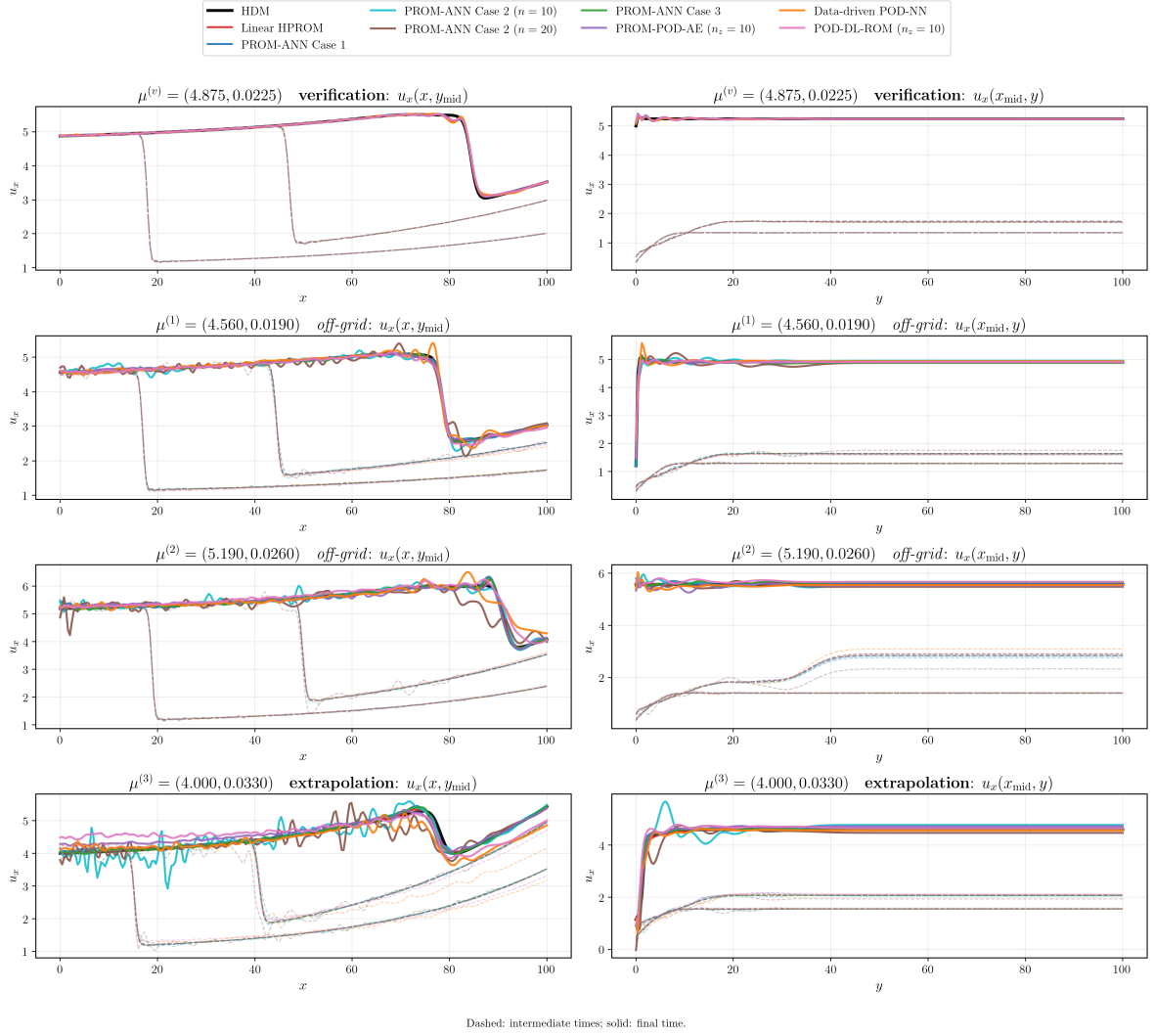


Figure 3: Current MLSPG-sensitive campaign: HDM, linear HPROM, selected PROM-ANN HPROMs, PROM-POD-AE with 2% ECSW, POD-NN-ROM, and POD-DL-ROM. Rows are ordered as verification point $\mu^{(v)}$, off-grid points $\mu^{(1)}$ and $\mu^{(2)}$, and extrapolatory point $\mu^{(3)}$.

10.4 Coefficient-Space Diagnostics

All coefficient-space diagnostics in this section use the MLSPG-sensitive linear HPROM coefficient trajectories as the reference:

$$\mathbf{Q}_{\text{ref}}(\mu) = \mathbf{Q}_{\text{lin}}^{\text{MLSPG-HPROM}}(\mu). \quad (49)$$

This is the relevant coefficient reference because the learned maps are trained from the same $\mathbf{V}_{\text{tot}}$ basis and from the same MLSPG-sensitive linear HPROM trajectories. For all completed models in this campaign, the coefficient diagnostics use the solver-side saved coordinates $\mathbf{q}_N(t)$ directly. No least-squares post-projection of state snapshots is used for the reported coefficient figures.

Figure 4 reports the per-index coefficient errors against this MLSPG-linear reference. The vertical dotted lines indicate the primary/secondary split for the two Case 2 configurations.

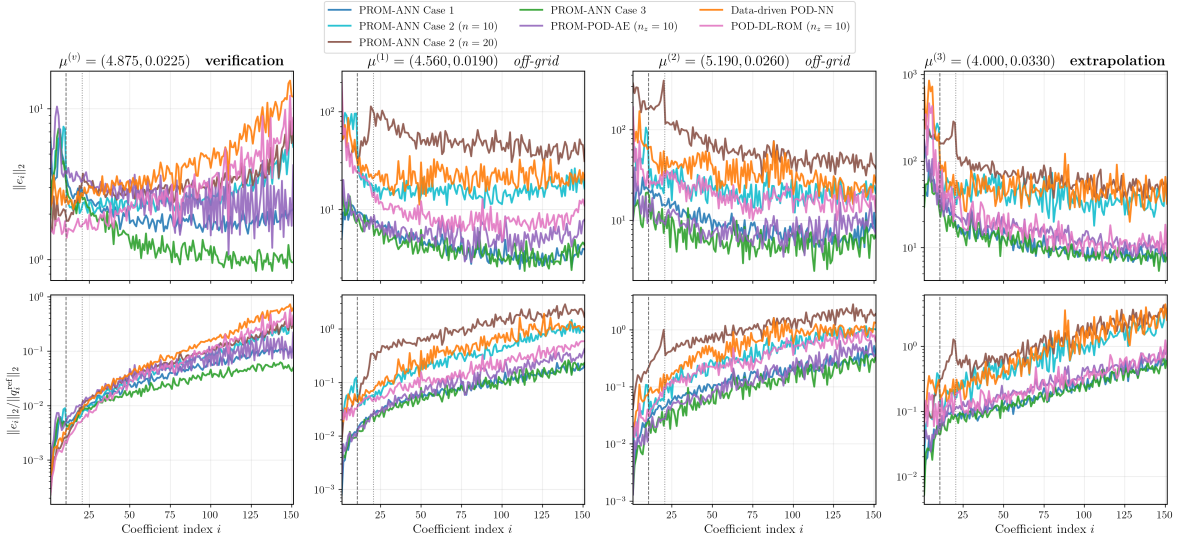


Figure 4: Current MLSPG-sensitive HPROM campaign: per-coefficient absolute and relative errors with respect to the MLSPG-sensitive linear HPROM coefficient trajectories. Panels are ordered as $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$, and $\mu^{(3)}$.

Figures 5 and 6 show the corresponding time-resolved coefficient-error heatmaps. The horizontal axis is time and the vertical axis is the coefficient index. The color scale is clipped at the 99th percentile to make the dominant error structures visible without being controlled by isolated outliers.

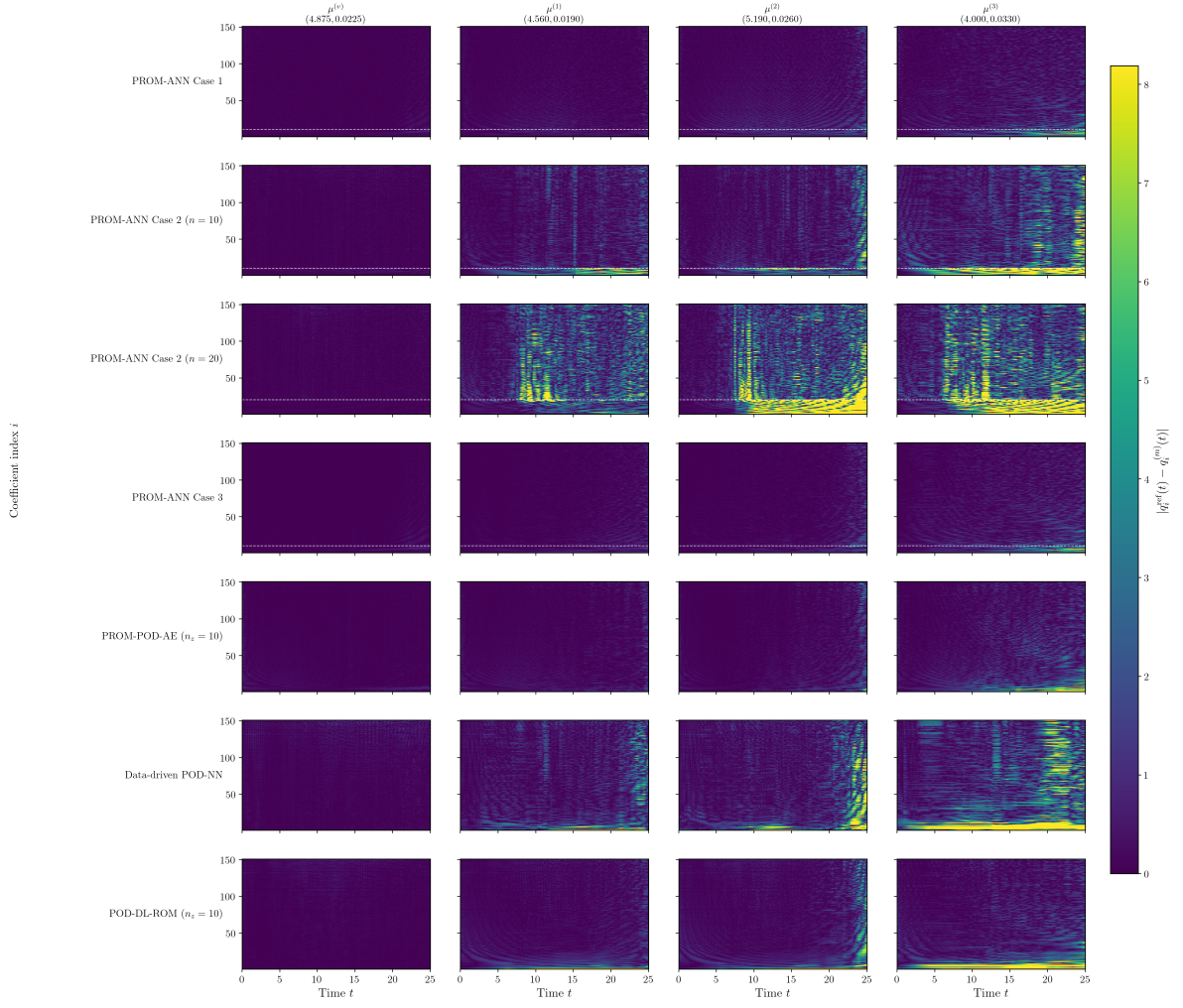


Figure 5: Current MLSPG-sensitive HPROM campaign: absolute coefficient-error heatmaps with respect to the MLSPG-sensitive linear HPROM coefficient trajectories. Columns are ordered as $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$, and $\mu^{(3)}$.

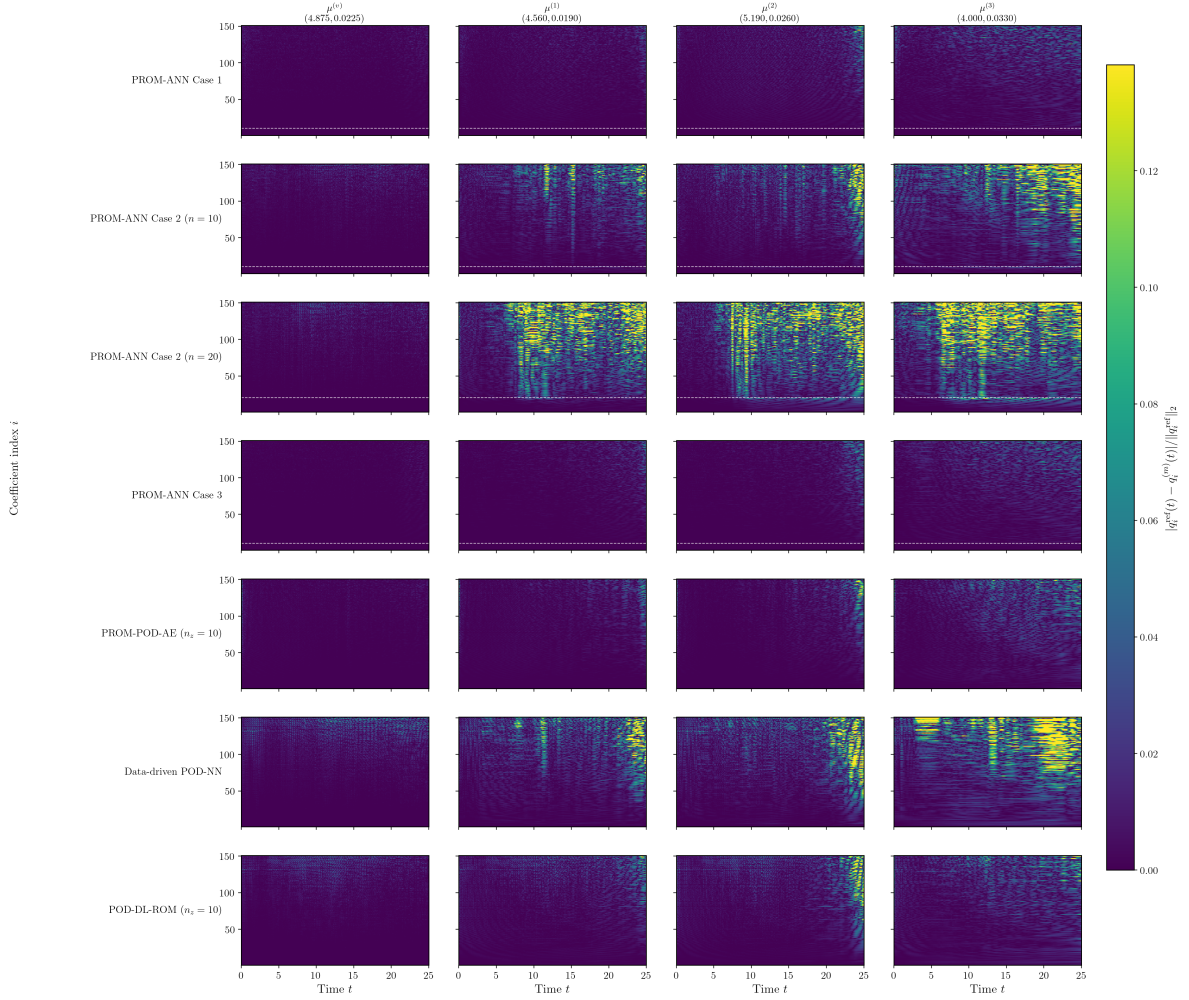


Figure 6: Current MLSPG-sensitive HPROM campaign: relative coefficient-error heatmaps with respect to the MLSPG-sensitive linear HPROM coefficient trajectories. Columns are ordered as $\boldsymbol{\mu}^{(v)}$, $\boldsymbol{\mu}^{(1)}$, $\boldsymbol{\mu}^{(2)}$, and $\boldsymbol{\mu}^{(3)}$.

10.5 Expanded Enriched MLSPG-Sensitive HPROM Campaign

The expanded enriched campaign repeats Stage 3 training and online evaluation after augmenting the baseline coefficient dataset with 36 additional linear-HPROM trajectories. The enrichment is deliberately restricted to the learning stage: the MLSPG-sensitive basis, HDM residual, time discretization, and fixed linear Stage-2 ECSW rule remain unchanged. The 36 enrichment points are generated by an LHS design with 18 samples inside $\mathcal{D}$ and 18 samples in the 25% expanded margin shown in Figure 2. The evaluation points, including $\boldsymbol{\mu}^{(3)}$, are excluded from the enrichment set. Thus, differences between the non-enriched and expanded-enriched rows isolate the effect of additional ROM-consistent coefficient data rather than a change in the trial space or in the linear hyperreduction rule.

All expanded-enriched coefficient outputs used below are solver-side saved $\mathbf{q}_N(t)$ arrays. No coefficient in the expanded-enriched tables or figures is obtained by least-squares post-projection of state snapshots. Table 7 reports the selected checkpoint for each family. The architecture choices are intentionally kept close to the non-enriched production campaign: the wide MLP widths are reused for the PROM-ANN and POD-NN-ROM maps, PROM-POD-AE uses a $n_z = 10$ GELU autoencoder, and POD-DL-ROM uses a $n_z = 10$ latent dynamics map.

The final PROM-ANN Case 1 checkpoint uses the same wide architecture but with GELU activation; a same-width SiLU candidate had a low offline validation error but was less robust in the intrusive online solve.

Table 7: Stage-3 network architectures used in the expanded enriched campaign. The dataset contains the baseline 9 trajectories plus 36 fixed-linear-HPROM enrichment trajectories.

Model	Learned map	Network architecture	Normalization / activation	Val. rel. Frobenius (%)	Trainable params
PROM-ANN Case 1	$\mathbf{q} \in \mathbb{R}^{10} \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{141}$	MLP hidden widths (256, 512, 512, 256)	z-score; GELU	0.365	564 621
PROM-ANN Case 2 ($n = 10$)	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{141}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	0.988	562 829
PROM-ANN Case 2 ($n = 20$)	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{131}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	1.511	560 259
PROM-ANN Case 3	$(\mathbf{q}, \mu_1, \mu_2, t) \in \mathbb{R}^{13} \mapsto \bar{\mathbf{q}} \in \mathbb{R}^{141}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	0.235	565 389
PROM-POD-AE	$\mathbf{q}_N \in \mathbb{R}^{151} \mapsto \mathbf{z} \in \mathbb{R}^{10} \mapsto \bar{\mathbf{q}}_N$	Encoder (512, 256, 128); decoder reverse; $n_z = 10$	z-score; GELU	0.077	486 817
POD-NN-ROM	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \mathbf{q}_N \in \mathbb{R}^{151}$	MLP hidden widths (256, 512, 512, 256)	z-score; SiLU	0.368	565 399
POD-DL-ROM	$(\mu_1, \mu_2, t) \in \mathbb{R}^3 \mapsto \mathbf{z} \in \mathbb{R}^{10} \mapsto \bar{\mathbf{q}}_N$	Encoder (512, 256); decoder (256, 512); dynamics (256, 512, 512, 256); $n_z = 10$	z-score; SiLU	0.158	952 747

Table 8 gives the expanded-enriched online errors. The fixed linear HPROM reference is repeated for orientation. The most important change relative to the non-enriched campaign is the stabilization of the learned maps on the extrapolatory point. At the extrapolatory point $\boldsymbol{\mu}^{(3)}$, POD-NN-ROM drops from 9.192% to 0.897%, POD-DL-ROM drops from 4.980% to 0.867%, Case 2 with $n = 10$ drops from 4.313% to 0.904%, and Case 2 with $n = 20$ drops from 6.440% to 0.914%. PROM-ANN Case 1 also improves at $\boldsymbol{\mu}^{(3)}$, from 1.563% to 0.847%, after selecting the GELU checkpoint; Case 3 improves from 1.438% to 0.848%, and PROM-POD-AE improves from 1.943% to 0.885%. On the original three evaluation points, POD-NN-ROM and POD-DL-ROM become close to the fixed linear HPROM, with mean errors 0.454% and 0.449%, respectively, compared with 0.448% for the linear HPROM. PROM-ANN Case 1 improves from a non-enriched mean of 0.582% to 0.461%, while the fragile Case 2 $n = 20$ row improves from 3.148% to 0.453%.

Table 8: Expanded enriched MLSPG-sensitive campaign: relative trajectory errors (%) with respect to HDM. The learned models are trained with the baseline 9 HPROM trajectories plus 36 additional fixed-linear-HPROM trajectories sampled in the original and 25% expanded parameter boxes.

Point	Linear HPROM	PROM-ANN Case 1	PROM-ANN Case 2 ($n = 10$)	PROM-ANN Case 2 ($n = 20$)	PROM-ANN Case 3	PROM-POD-AE	POD-NN-ROM	POD-DL-ROM
$\boldsymbol{\mu}^{(e)}$	0.413	0.413	0.418	0.417	0.429	0.458	0.419	0.415
$\boldsymbol{\mu}^{(1)}$	0.460	0.485	0.465	0.465	0.509	0.593	0.463	0.460
$\boldsymbol{\mu}^{(2)}$	0.472	0.484	0.483	0.476	0.473	0.491	0.481	0.473
$\boldsymbol{\mu}^{(3)}$	0.868	0.847	0.904	0.914	0.848	0.885	0.897	0.867

Table 9 compares the expanded enriched campaign directly with the non-enriched baseline. The table separates the mean over the original three points from the extrapolatory stress-test point. This split is important because the expanded LHS design is not merely a refinement of the original box; it is also intended to provide coefficient data near and outside the original parameter boundary.

Table 9: Effect of the expanded 36-trajectory enrichment relative to the non-enriched baseline. Errors are relative trajectory errors (%) versus HDM. Negative changes indicate improvement after enrichment. The first three columns average over $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$; the last three isolate the extrapolatory point $\mu^{(3)}$.

Method	Baseline mean (3 pts.)	Ext. enrich. mean (3 pts.)	Change	Baseline $\mu^{(3)}$	Ext. enrich. $\mu^{(3)}$	Change
Linear HPROM	0.448	0.448	+0.000	0.868	0.868	+0.000
PROM-ANN Case 1	0.582	0.461	-0.121	1.563	0.847	-0.716
PROM-ANN Case 2 ($n = 10$)	1.270	0.455	-0.815	4.313	0.904	-3.409
PROM-ANN Case 2 ($n = 20$)	3.148	0.453	-2.695	6.440	0.914	-5.526
PROM-ANN Case 3	0.515	0.471	-0.044	1.438	0.848	-0.590
PROM-POD-AE	0.594	0.514	-0.080	1.943	0.885	-1.057
POD-NN-ROM	1.601	0.454	-1.147	9.192	0.897	-8.294
POD-DL-ROM	1.161	0.449	-0.712	4.980	0.867	-4.113

The expanded enrichment improves every learned row in the aggregate comparison of Table 9, but the magnitude of the improvement is strongly model dependent. The direct non-intrusive maps benefit most from the expanded coverage, as expected for maps driven only by (μ, t) . PROM-ANN Case 2 also improves substantially, especially for the $n = 20$ split that was fragile in the non-enriched campaign. After enrichment, Case 1, Case 3, and the two Case 2 splits all lie close to the fixed linear HPROM on the original three-point mean, while PROM-POD-AE remains slightly less accurate on the in-domain off-grid points but gains clearly at $\mu^{(3)}$. The Case 1 activation diagnostic indicates that offline closure accuracy alone is not a sufficient selection criterion for intrusive closure models: the final GELU checkpoint is retained because it preserves the low offline error while producing a more stable online LSPG trajectory than the same-width SiLU candidate.

The online dimensions, ECSW mesh sizes, and timings are reported in Table 10. The learned intrusive models use the same 2% ECSW convention as the baseline production tables. In this expanded campaign, Case 1, Case 2 with $n = 10$, Case 3, and PROM-POD-AE retain 901 ECSW elements, while Case 2 with $n = 20$ retains 1801 elements. The linear HPROM row uses the fixed Stage-2 ECSW rule with 5975 retained elements. Non-intrusive inference timings are CPU timings from saved run summaries and should be read as conservative online costs.

Table 10: Expanded enriched MLSPG-sensitive campaign: online dimensions, ECSW mesh sizes, average online timings over the four evaluation points, and speedups with respect to the HDM mean time $t_{\text{HDM}} = 737.44$ s. Learned intrusive models use selected 2% ECSW rules, the linear HPROM uses the fixed Stage-2 ECSW rule, and non-intrusive models use direct network inference with no ECSW.

Method	Online/latent dim.	$\bar{n}$	N_e	Mean online time (s)	Mean speedup vs HDM
Linear HPROM	151	–	5975	145.2091	5.1
PROM-ANN Case 1	10	141	901	9.2240	79.9
PROM-ANN Case 2 ($n = 10$)	10	141	901	2.6048	283.1
PROM-ANN Case 2 ($n = 20$)	20	131	1801	3.9657	186.0
PROM-ANN Case 3	10	141	901	9.2615	79.6
PROM-POD-AE	10	–	901	8.4468	87.3
POD-NN-ROM	151	–	–	0.0113	65171.1
POD-DL-ROM	10	–	–	0.0213	34630.0

Figure 7 compares the expanded enriched models against HDM and the fixed linear HPROM at the four reported evaluation points, including the extrapolatory stress-test point $\mu^{(3)}$.

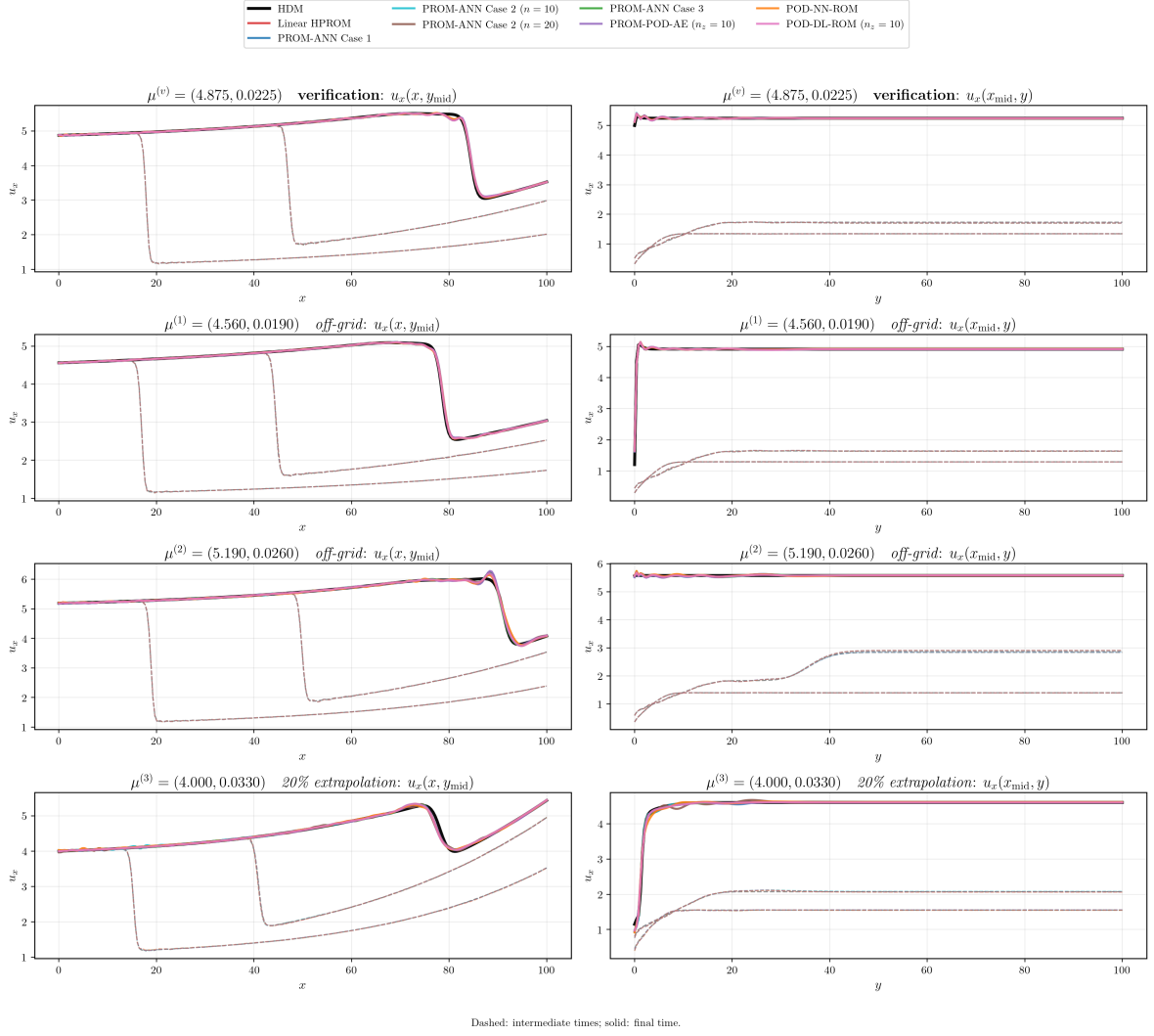


Figure 7: Expanded enriched MLSPG-sensitive campaign: HDM, fixed linear HPROM, expanded-enriched PROM-ANN HPROMs, expanded-enriched PROM-POD-AE with 2% ECSW, expanded-enriched POD-NN-ROM, and expanded-enriched POD-DL-ROM. ROM curves are reconstructed from saved solver-side snapshots. Rows are ordered as $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$, and $\mu^{(3)}$.

The expanded-enriched coefficient diagnostics in Figures 8–10 use the fixed MLSPG-sensitive linear HPROM coefficient trajectories as the reference at all four evaluation points. These diagnostics show the same qualitative picture as the state errors: POD-NN-ROM and POD-DL-ROM become close to the linear coefficient trajectories across the expanded test set, Case 2 benefits from the additional parameter coverage, and Case 1 remains stable after switching to the final GELU checkpoint.

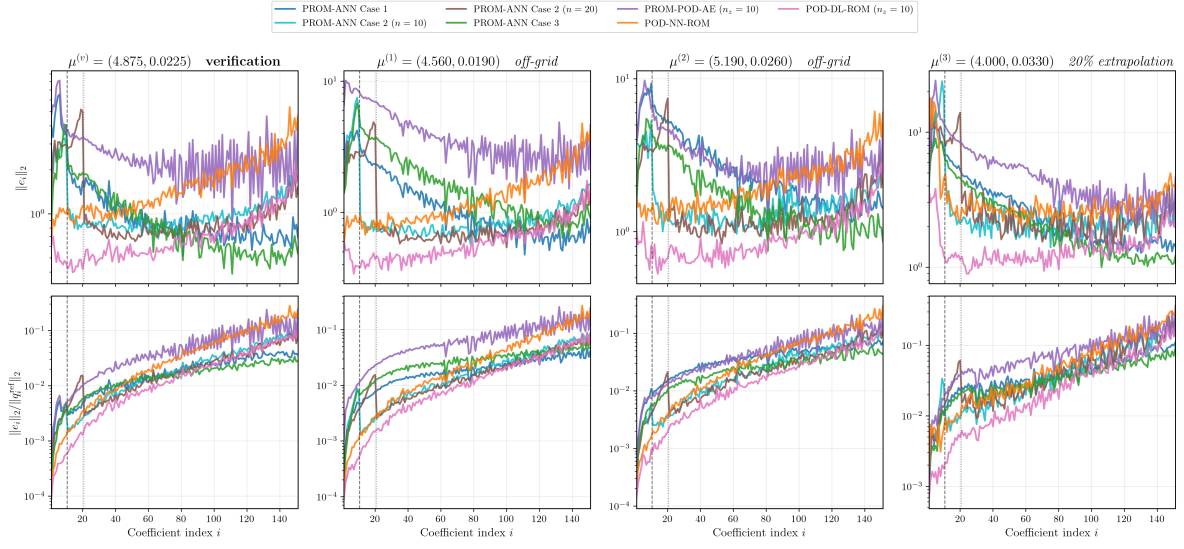


Figure 8: Expanded enriched MLSPG-sensitive campaign: per-coefficient absolute and relative errors with respect to the fixed MLSPG-sensitive linear HPROM coefficient trajectories. Panels are ordered as $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$, and $\mu^{(3)}$.

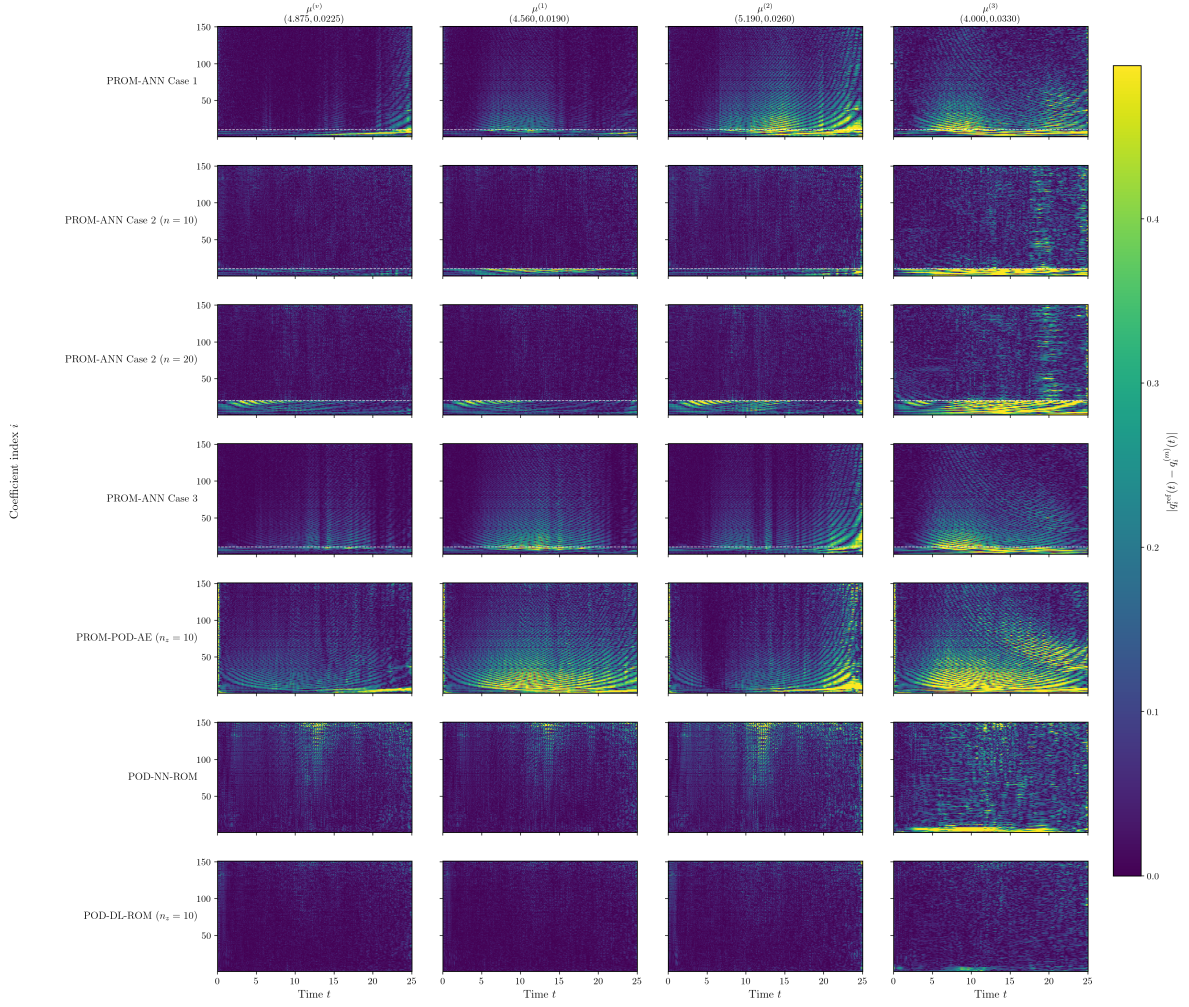


Figure 9: Expanded enriched MLSPG-sensitive campaign: absolute coefficient-error heatmaps with respect to the fixed MLSPG-sensitive linear HPRM coefficient trajectories. Columns are ordered as $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$, and $\mu^{(3)}$.

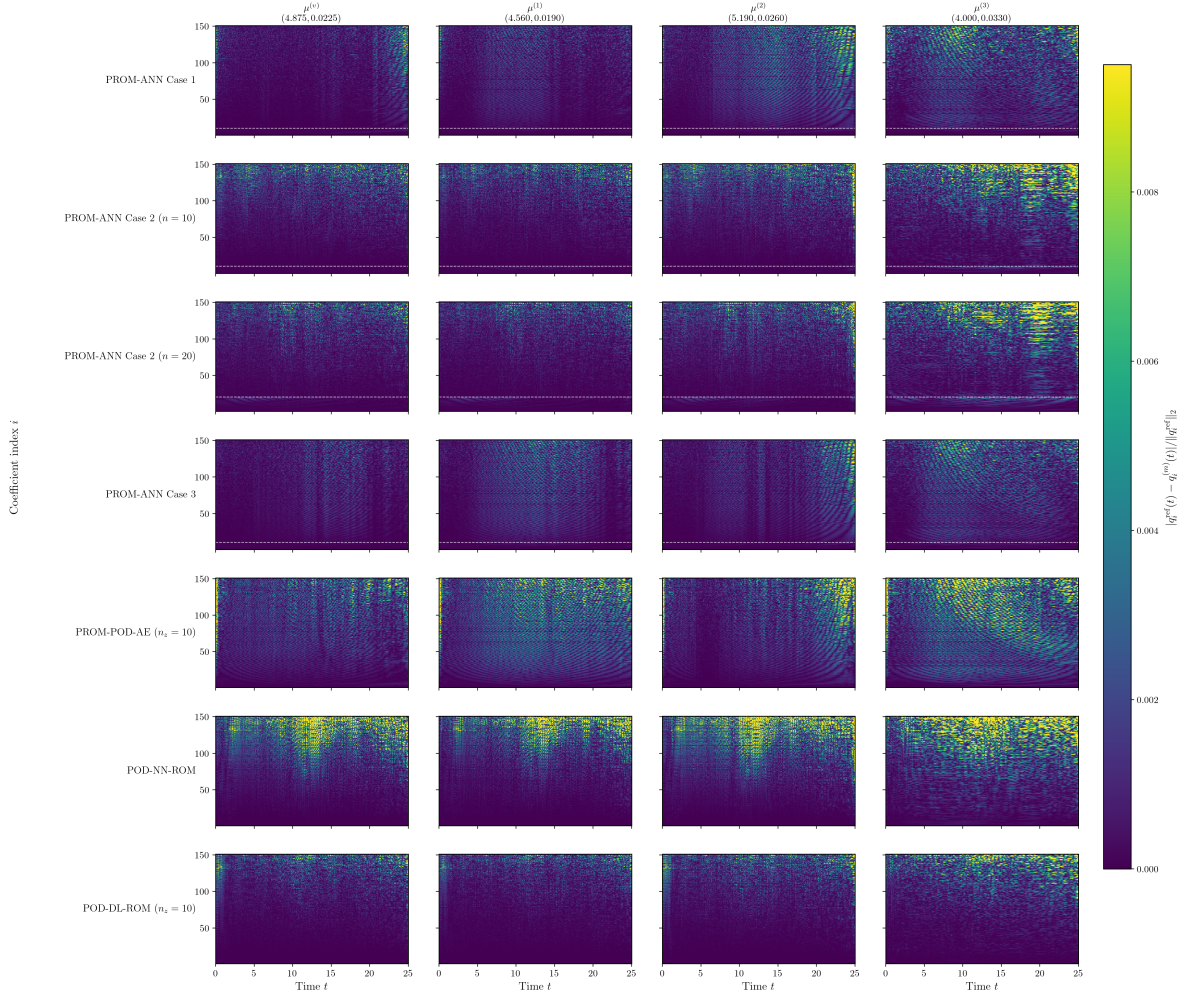


Figure 10: Expanded enriched MLSPG-sensitive campaign: relative coefficient-error heatmaps with respect to the fixed MLSPG-sensitive linear HPROM coefficient trajectories. Columns are ordered as $\boldsymbol{\mu}^{(v)}$, $\boldsymbol{\mu}^{(1)}$, $\boldsymbol{\mu}^{(2)}$, and $\boldsymbol{\mu}^{(3)}$.

11 Shallow-Water Benchmark and Experimental Protocol

11.1 Governing equations and parameterized initial condition

The second benchmark is a 2D shallow-water model on a flat bottom, in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \quad \mathbf{U} = \begin{bmatrix} h \\ hu_x \\ hu_y \end{bmatrix}, \quad (50)$$

with

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} hu_x \\ hu_x^2 + \frac{1}{2}gh^2 \\ hu_xu_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} hu_y \\ hu_xu_y \\ hu_y^2 + \frac{1}{2}gh^2 \end{bmatrix}. \quad (51)$$

The parametric scenario is a collision of two Gaussian bumps in water depth:

$$h(x, y, 0; \boldsymbol{\mu}) = h_0 + \mu_1 \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_1\|_2^2}{2\sigma^2}\right) + \mu_2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_2\|_2^2}{2\sigma^2}\right), \quad (52)$$

with zero initial velocity components ($u_x = u_y = 0$), $h_0 = 1$, $\sigma = 0.045$, $\mathbf{c}_1 = (0.30, 0.50)$, $\mathbf{c}_2 = (0.70, 0.50)$, and

$$\boldsymbol{\mu} = (\mu_1, \mu_2) \in [0.06, 0.14] \times [0.06, 0.14]. \quad (53)$$

Reflective boundary conditions are imposed on all sides.

11.2 Numerical setup

The domain $\Omega = [0, 1] \times [0, 1]$ is discretized with $N_x = N_y = 600$, leading to $N = 3N_xN_y$ unknowns. The HDM uses MUSCL reconstruction with HLL/HLLC interface fluxes and an implicit BDF2 time integrator (BE startup). We use a fixed time step $\Delta t = 7 \times 10^{-5}$ up to $T_f = 0.28$, yielding 4001 stored states per parameter value.

11.3 Sampling protocol aligned with Burgers

The staged workflow mirrors the Burgers protocol: baseline set of 9 structured parameter points (3×3 grid), an HPROM-generated LHS enrichment set (36 points in the expanded Burgers campaign), Stage 1 POD basis $\mathbf{V}_{\text{tot}}$ from HDM snapshots, Stage 2 ROM-consistent $\mathbf{q}_N$ dataset from linear HPROM in $\mathbf{V}_{\text{tot}}$, Stage 3 training of PROM-ANN (Cases 1–3), PROM-POD-AE, POD-NN-ROM, and POD-DL-ROM, followed by online intrusive evaluation with HPROM settings. This protocol preserves methodological consistency across both benchmarks, so that observed performance differences are attributable to model structure and problem physics rather than to workflow changes.

12 Scope of the Study

Core model families are the linear HPROM reference, PROM-ANN (Cases 1–3), PROM-POD-AE, POD-NN-ROM, and POD-DL-ROM. The baseline/enriched comparison is retained as a controlled data axis. PG/PRNN-style variants are treated as optional diagnostic tracks and are not part of the core comparative tables in this manuscript.

A Burgers Basis-Selection Diagnostics

The M -regularized LSPG-sensitive basis was selected through two diagnostics. The first diagnostic is the low/high transfer operator $\mathbf{T}_{LH}$ introduced in (17). Table 11 reports the singular-value statistics of this operator over the three Burgers evaluation trajectories. The LSPG-sensitive basis reduces the mean value of $\sigma_{\max}(\mathbf{T}_{LH})$ by approximately one half relative to the Euclidean POD basis, and also reduces the 95th percentile and maximum values. This supports the use of the modified metric as a robustness device for intrusive nonlinear reduced models: if the nonlinear closure, decoder, or surrogate produces errors in the unresolved part of the representation, the resulting feedback into the resolved coordinates through the LSPG normal equations is less amplifying.

Table 11: Burgers low/high transfer diagnostic for basis selection. The table reports statistics of $\sigma_{\max}(\mathbf{T}_{LH})$ and of the corresponding M -weighted state gain g_M over the original three evaluation trajectories.

Basis	mean $\sigma_{\max}(T_{LH})$	p95 $\sigma_{\max}(T_{LH})$	max $\sigma_{\max}(T_{LH})$	mean g_M	p95 g_M	max g_M
Euclidean POD	3.078e-02	3.668e-02	4.088e-02	3.078e-02	3.668e-02	4.088e-02
LSPG-sensitive POD	1.548e-02	2.244e-02	2.433e-02	1.599e-02	2.339e-02	2.559e-02

As a complementary check, we perturb only the high-coordinate block of an oracle reconstruction and measure the induced deviation from the linear reduced trajectory. This test

is implemented with the Case 2 partition because it isolates the most direct contamination mechanism: the high block is injected independently of the online low-coordinate solve. Table 12 reports averages over the original verification point, two off-grid points, and five random seeds for the perturbed cases. The M -regularized LSPG-sensitive basis slightly increases the unperturbed linear error in this preliminary test, but reduces the high-to-low contamination measured against the linear reference, especially in the resolved coordinates. The intended role of the metric is broader than this specific test: it provides a protective basis choice for nonlinear intrusive models in which learned high coordinates or decoded manifold corrections may contain approximation error. This diagnostic is not used as a standalone performance result; it supports the basis choice used in the main Burgers campaign.

Table 12: Oracle high-mode perturbation diagnostic for basis selection. Errors are averaged over $\mu^{(v)}$, $\mu^{(1)}$, $\mu^{(2)}$; perturbed rows also average over five random seeds.

Basis metric	Unperturbed ε_u vs HDM (%)	1% pert., ε_u vs linear (%)	1% pert., ε_q^{low} vs linear (%)	2% pert., ε_u vs linear (%)	2% pert., ε_q^{low} vs linear (%)
Euclidean POD	0.441	0.091	0.014	0.183	0.045
M -regularized LSPG-sensitive POD	0.448	0.088	0.012	0.177	0.035

B Case 2 Trimmed-Closure Diagnostic

The non-enriched results show that PROM-ANN Case 2 with $n = 10$ is more robust than the independently trained Case 2 model with $n = 20$, especially away from the verification point. To test whether this behavior is caused primarily by the larger online split or by the learned parameter-time tail map itself, we ran an additional diagnostic. The diagnostic reuses the trained $n = 10$ Case 2 map

$$(\mu_1, \mu_2, t) \mapsto (q_{11}, \dots, q_{151}), \quad (54)$$

discards its first ten secondary outputs, and injects only $(q_{21}, \dots, q_{151})$ in an $n = 20$ online HPROM solve. Thus, the online unknown is enlarged from 10 to 20 primary coordinates, but the high-mode closure is inherited from the better-performing $n = 10$ map rather than from an independently trained $n = 20$ network. The same diagnostic is also run at the extrapolatory point $\mu^{(3)}$ introduced in the main text. This is a diagnostic variant only; it is not included in the main selected-model comparison.

Table 13: Case 2 diagnostic in the non-enriched MLSPG-sensitive campaign. The trimmed variant uses the trained $n = 10$ Case 2 map $(\mu_1, \mu_2, t) \mapsto (q_{11}, \dots, q_{151})$ and discards its first ten secondary outputs, so that the online solve uses $n = 20$ primary coordinates and injects only $(q_{21}, \dots, q_{151})$. Errors are relative trajectory errors (%) with respect to HDM. The in-domain mean averages $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$; the extrapolatory point $\mu^{(3)}$ is reported separately. Timings are averages over all four displayed points.

Method	$E_{\mu^{(v)}}$	$E_{\mu^{(1)}}$	$E_{\mu^{(2)}}$	In-domain mean E	$E_{\mu^{(3)}}$	Mean online time (s)	n_e
Native Case 2 ($n = 10$)	0.437	1.808	1.565	1.270	4.313	2.6251	901
Native Case 2 ($n = 20$)	0.441	3.433	5.569	3.148	6.440	3.9693	1801
Trimmed $n = 20$ from $n = 10$ map	—	—	—	—	—	—	—

Figure 11 visualizes the same comparison. The trimmed run substantially reduces the degradation of the native $n = 20$ Case 2 model at the two off-grid points and at $\mu^{(3)}$, which indicates that the independently trained $n = 20$ parameter-time closure is the main source of the poor generalization. However, the trimmed run is not uniformly better than the native $n = 10$ solve on the original in-domain points and is more expensive because the online unknown dimension is larger and the diagnostic ECSW rule retains more elements. This supports keeping the trimmed model in the appendix rather than promoting it to the main baseline table.

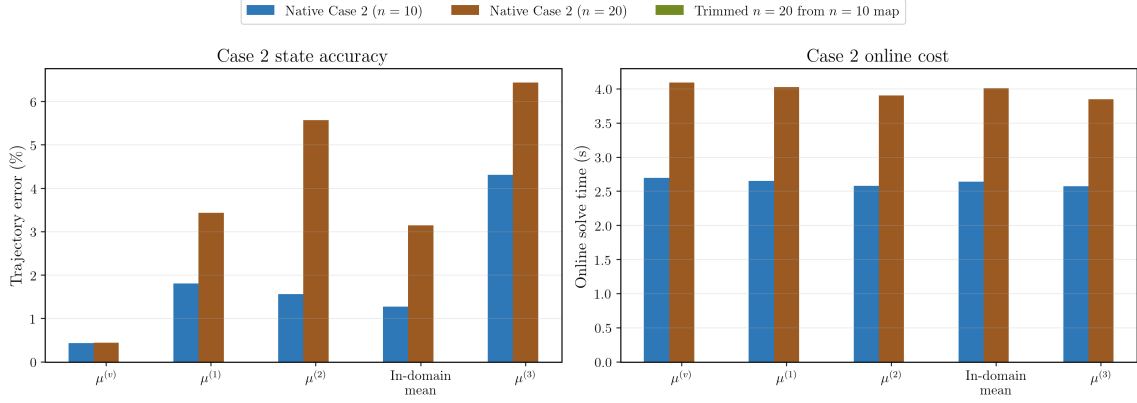


Figure 11: Case 2 trimmed-closure diagnostic in the non-enriched MLSPG-sensitive campaign. The trimmed variant uses an $n = 20$ online solve but reuses the trained $n = 10$ Case 2 map after discarding the first ten predicted secondary coordinates. The extrapolatory point $\mu^{(3)}$ is reported separately from the in-domain mean.

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